Vehicle Analytics System - Testing & User Guide

Overview

The **Vehicle Analytics System** is a Django-based machine learning web application that provides actionable insights into the vehicle market. It utilizes a dummy dataset of vehicle sales to perform Exploratory Data Analysis (EDA) and run predictive models.

The system incorporates three core machine learning functionalities:

1. **Regression Analysis (Random Forest Regressor):** Predicts the estimated selling price of a vehicle based on its manufacturing year, kilometers driven, seating capacity, and the owner's estimated income.
2. **Classification Analysis (Random Forest Classifier):** Categorizes the income level (e.g., Low, Medium, High) of a vehicle owner based on vehicle specifications.
3. **Clustering Analysis (K-Means Clustering):** Segments clients into groups ("Economy", "Standard", "Premium") based on income and estimated vehicle price. It inherently combines the regression output for price prediction to plot the final cluster.

How to Start the Application

To test the application, you first need to ensure the Django development server is running.

1. Open a terminal.
2. Navigate to the project directory:

   ```
   cd ~/Downloads/django_ml_project
   ```

3. Activate the Python virtual environment:

   ```
   source venv/bin/activate
   ```

4. Start the Django server:

   ```
   python manage.py runserver
   ```

5. Open a web browser and go to: `http://127.0.0.1:8000/`

How to Test the Application

Once the application is loaded in your browser, you can navigate using the left sidebar to test different modules.

1. Exploratory Data Analysis (EDA)

- **What it does:** Displays the raw dataset, basic statistical summaries, and a Plotly

geographic map showing the concentration of vehicle clients across Rwanda's districts (Exercise 19a).

- **How to test:**
  - Click on "Data Exploration" in the sidebar.
  - Scroll through the Data Table and Statistical Analysis sections to ensure data is loading correctly.
  - View the "Geographic Distribution" section to confirm the Plotly map renders interactively. You can hover, zoom, and pan around the map.

2. Regression Analysis

- **What it does:** Predicts a vehicle's selling price.
- **How to test:**
  - Click on "Regression Analysis" in the sidebar.
  - In the "Input Vehicle Specifications" form, enter some test values (e.g., Year: 2022, Kilometers: 15000, Seats: 5, Income: 60000).
  - Click the "**Predict Market Price**" button.
  - **Expected Result:** The right-hand panel should update from "Waiting..." to display an Estimated Value in dollars. Below, verify that the R^2 Evaluation Metrics are visible.

3. Classification Analysis

- **What it does:** Predicts a vehicle owner's internal income category label.
- **How to test:**
  - Click on "Classification Analysis" in the sidebar.
  - Fill out the form with sample inputs (e.g., Year: 2018, Kilometers: 50000, Seats: 5, Income: 20000).
  - Click the "**Predict Income Category**" button.
  - **Expected Result:** The right-hand panel should update to show the Predicted Category (e.g., "Medium"). Check the Evaluation Metrics section below for model accuracy and test set comparisons.

4. Clustering Analysis & Client Segmentation

- **What it does:** Uses a combination of Regression (to predict selling price) and K-Means clustering to place the client into a specific market segment ("Economy", "Standard", "Premium"). It also highlights evaluation metrics heavily refined (Silhouette Score > 0.9) and variance calculations (Exercise 19b).
- **How to test:**
  - Click on "Clustering Analysis" in the sidebar.
  - Enter test data (e.g., Year: 2024, Kilometers: 1000, Seats: 5, Income: 150000).
  - Click "**Run Combined Inference**".
  - **Expected Result:** The dual-model engine will predict both the value of the vehicle and the assigned Client Cluster badge (e.g., "Premium").
  - Scroll down to "Evaluation Metrics" to verify that the **Silhouette Score** is prominently displayed and properly refined, along with the coefficient of variation details.

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